

Business Problem Statement

A real estate consulting and investment firm seeks to gain deeper insights into housing market dynamics to support data-driven investment, pricing, and market expansion decisions. The company has observed significant variations in property prices, sales activity, and buyer negotiation behavior across different cities and regions. Additionally, changing economic conditions such as inflation and interest rates may be influencing property demand and valuation.

The firm wants to understand how property characteristics, including house type, size, number of rooms, construction period, and location, affect market value and purchasing trends. They are also interested in identifying high-performing regions, emerging real estate markets, and factors that contribute to higher property valuations.

Using historical property transaction data along with macroeconomic indicators, this project aims to analyze market trends, evaluate regional performance, and uncover key drivers of property prices.

The overarching business question is:

“How can real estate transaction and economic data be leveraged to identify market trends, evaluate investment opportunities, optimize pricing strategies, and support data-driven decision-making in the housing market?”

Key objectives include:

- Identifying the cities and regions with the highest property values and strongest market performance.
- Understanding the impact of property characteristics such as house type, size, number of rooms, and build era on purchase prices.
- Analyzing how inflation and interest rates influence housing prices and market activity.
- Evaluating buyer negotiation behavior across different regions and property segments.
- Discovering emerging markets and potential investment opportunities through geographic and trend analysis.
- Developing an interactive analytics dashboard to support strategic business and investment decisions.